MIL-B-7370B
10 Sep 1969
SUPERSEDING
MIL-B-7370A
12 June 1953

MILITARY SPECIFICATION

BULB, TEMPERATURE, THERMORESISTOR, -70° TO +150°C, TYPE G-1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of multitemperature, electrical-resistance bulb, designated type G-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated List
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards, Order Of Precedence For The Selection Of

FSC 6685

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MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS33678	Connector, Receptacle, Electrical, Ingegral Mounting
MS35769	Gasket, Metallic-Encased, Annular Copper And Asbestos

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations

(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

3. REQUIREMENTS

* 3.1 Qualification testing. This specification makes provisions for qualification testing.

* 3.2 Qualification. Bulbs furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Materials.

* 3.3.1 Metals. Metals shall be corrosion-resistant unless protected to resist corrosion during normal service life. The use of dissimilar metals, especially brass, copper, or steel in intimate metal-to-metal contact with aluminum or aluminum alloy shall be avoided where practicable.

* 3.3.2 Steel or brass. Steel or brass parts shall be suitably protected to meet the tests specified herein.

* 3.3.3 Protective treatment. When materials are used in the construction of the bulb that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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* 3.3.4 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

* 3.3.4.1 Standard parts. Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

* 3.4 Design. The resistance bulb shall conform to Figure 1 and shall be designed for high-frequency vibration application.

* 3.4.1 Temperature range. The bulb shall be designed to function satisfactorily throughout the temperature range of -70° to +150° Centigrade (C).

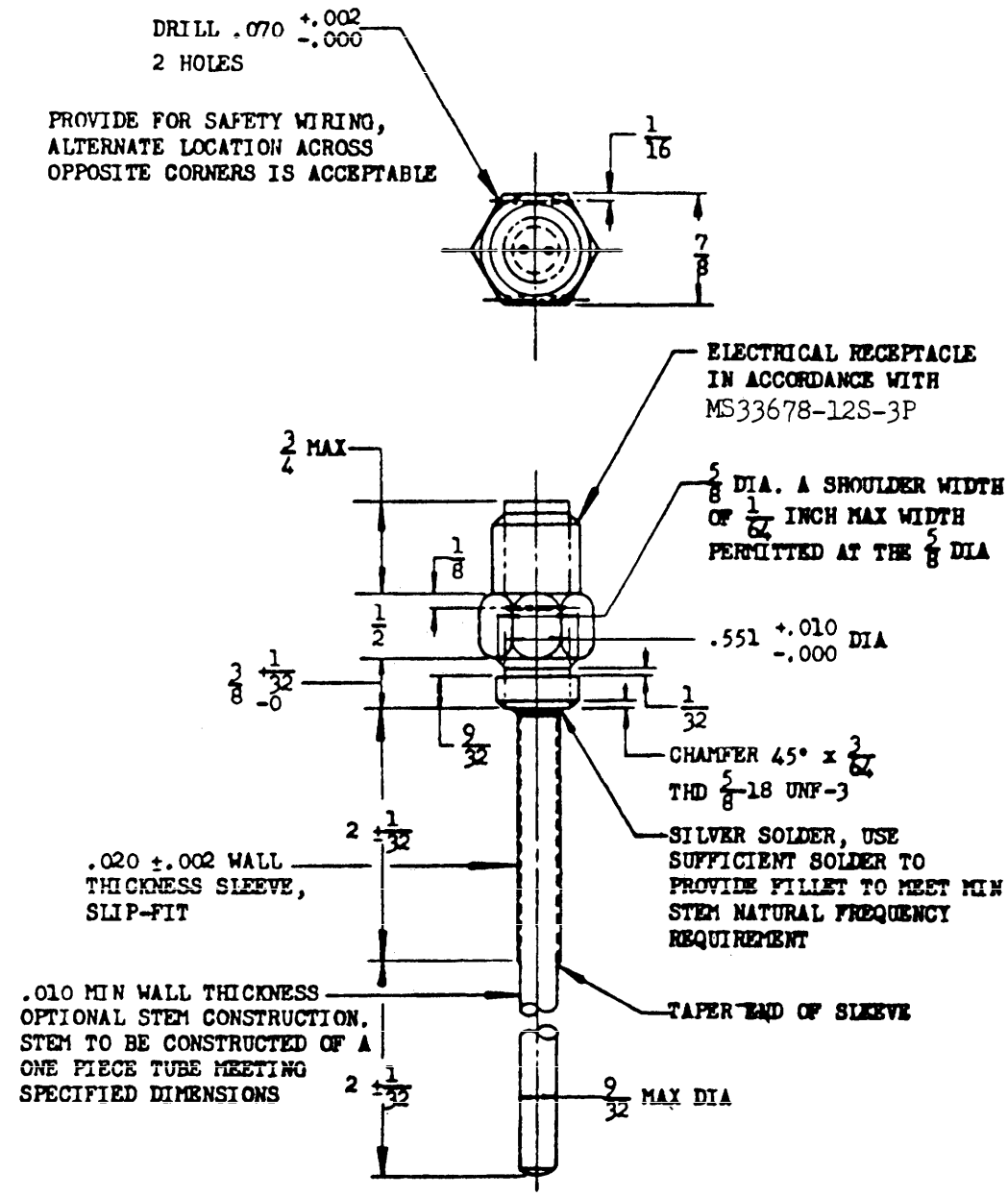
* 3.4.2 Temperature-resistance characteristics. The temperature-resistance characteristics of the bulb shall conform to the values specified in Table I.

TABLE I

Temperature-Resistance Characteristics

Temperature °Centigrade	Resistance-ohms	Temperature °Centigrade	Resistance-ohms
-70	68.27 ±0.50	+50	108.39 ±0.40
-60	71.19 ±0.50	+60	112.28 ±0.50
-50	74.24 ±0.40	+70	116.27 ±0.50
-40	77.39 ±0.40	+80	120.36 ±0.50
-30	80.56 ±0.40	+90	124.55 ±0.50
-20	83.77 ±0.40	+100	128.85 ±0.50
-10	87.04 ±0.40	+110	133.26 ±0.50
0	90.38 ±0.40	+120	137.78 ±0.60
+10	93.80 ±0.40	+130	142.40 ±0.60
+20	97.31 ±0.40	+140	147.11 ±0.60
-30	100.91 ±0.40	+150	151.91 ±0.60
+40	104.60 ±0.40	- -	- - - - - -

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±2°, FRACTIONS ±1/64.

FIGURE 1. RESISTANCE BULB

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* 3.4.3 Resistance element. The resistance element of the bulb shall be of nickel winding and shall be hermetically sealed in a metal well.

* 3.4.3.1 The resistance winding shall be evenly distributed over the bulb core and shall cover a minimum of 3-1/4 inches of the stem length.

* 3.5 Construction. The bulb shall be so constructed as to withstand without failure, the normal strains, jars, vibrations, and such other conditions that are incident to shipping, storage, installation, and service. The stem of the bulb shall be so brazed to the bulb head as to prevent any leakage into the bulb winding or electrical receptacle.

* 3.5.1 Gasket. An MS35769-10 gasket shall be furnished with each bulb.

* 3.6 Performance. The bulb shall be capable of meeting the requirements specified in Section 4, when subjected to the applicable tests.

* 3.7 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

* 3.8 Finish. Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of temperature and atmospheric conditions shall not be used.

* 3.9 Screw threads. Screw threads shall be in accordance with MIL-S-7742.

* 3.10 Weight. The weight of the bulb shall not exceed 0.30 pound.

* 3.11 Identification of product. The bulb shall be marked in accordance with MIL-STD-130. The following information shall be legibly stamped or engraved on any one of the faces of the hexagonal head of the bulb.

Type G-1
Manufacturer's Part No.
Manufacturer's Serial No.
Manufacturer's name or trade-mark

* 3.11.1 Use of AN or MIL designations. AN or MIL designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

3.12 Workmanship.

* 3.12.1 General. The bulb, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to soldering, wiring, marking of parts and assemblies and freedom of parts from burrs and sharp edges.

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* 3.12.2 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the bulbs, they shall be held or limited accordingly.

* 3.12.3 Cleaning. The bulbs shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material prior to final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of bulbs shall be classified as follows:

a. Qualification tests. Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

b. Quality conformance test. Quality conformance test is that test accomplished on bulbs manufactured and submitted for acceptance under contract.

4.3 Test conditions.

* 4.3.1 Standard atmospheric conditions. Whenever the pressure and temperatures existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches mercury) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

* 4.3.2 Vibration. A vibration stand shall be used which will vibrate at any desired frequency between 0 and 1,000 cycles per second and shall subject the bulb to a simple harmonic motion at the amplitude specified herein.

* 4.3.3 Resistance values. Where practicable, the resistance values specified for test results shall be the averages of direct and reversed readings, to eliminate thermoelectric potential errors.

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* 4.3.4 Test voltage. Unless otherwise specified, a constant voltage of 1.5 ±0.2 volt direct current, shall be applied across the bulb for making resistance measurements.

4.4 Qualification tests.

* 4.4.1 Prior qualification. Unless otherwise specified by the procuring activity, bulbs which have not previously passed a qualification test, or which have passed the qualification tests and have been modified in any manner, shall satisfactorily pass a qualification test prior to the acceptance of any bulb.

* 4.4.2 Sampling instructions. The qualification tests samples shall consist of three bulbs upon which qualification is desired. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from the qualifying activity. (See 6.3)

* 4.4.3 Tests. The qualification tests shall consist of all the tests specified under quality conformance tests.

* 4.5 Quality conformance tests. Quality conformance tests shall consist of individual tests, sampling tests, and special tests.

* 4.5.1 Individual tests. Each bulb shall be subjected to the following tests in the order listed.

* 4.5.1.1 Examination of product. The bulb shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

* 4.5.1.2 Bulb adjustment. The resistance of the bulb shall be measured at 0°C. The bulb shall be immersed in an ice bath to the top of the attaching threads for conducting this test. The measured resistance of the bulb shall conform to the value specified in Table I.

* 4.5.1.3 Vibration (natural frequency). The bulb shall be mounted by the attaching threads, on a suitably designed vibration stand as specified herein. No additional support of the stem shall be made other than that provided by the attaching threads. The simple harmonic motion shall be applied perpendicular to the longitudinal axis of the bulb stem. The frequency shall be varied from 0 to 700 cycles per second at low acceleration (maximum of 0.003 inch amplitude). The natural frequency of the bulb stem shall exceed 700 cycles per second. The bulb resistance shall be monitored throughout the test and intermittent element resistance, or insulation resistance of the element to the housing of less than 20 megohms, shall be cause for rejection. At the conclusion of the test the resistance element shall meet the requirements of Table I at any test point(s) chosen by the inspector. The insulation resistance of the element to the housing shall not be less than 20 megohms.

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* 4.5.2 Sampling tests. Five bulbs shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following sampling tests. These tests shall be in addition to the individual tests specified herein.

* 4.5.2.1 Bulb resistance. The bulb resistance of each bulb shall be measured at temperature of -50° ±5°C, and +120° ±5°C. The measured resistance shall be within the tolerance specified in Table I for the exact temperature of the master. To determine the true resistance equivalent to the temperature of the master, interpolation of the values specified in Table I will be required. The bulb shall be immersed in the agitated liquid baths to the top of the attaching threads for all resistance measurements.

* 4.5.3 Special tests. Three bulbs selected at random from the first 15 bulbs representing the manufacturer's contract or order shall be subjected to any or all of the following special tests specified herein.

* 4.5.3.1 Detail examination. The bulb shall be critically examined to determine full compliance with regard to weight, physical dimensions, markings, and similar detail requirements.

* 4.5.3.2 Response time. The sensitive element of the bulb shall be quenched from an agitated liquid bath of 100°C to an agitated ice bath at 0°C. The time required for the bulb temperature to fall from 100° to 10°C shall be noted. In conducting this test, a calibrated indicator may be connected to the bulb or a Wheatstone bridge may be used. The time required shall not exceed 8 seconds.

* 4.5.3.3 Thermal conductivity. The head of the bulb shall be packed in ice providing a head temperature of substantially 0°C. The stem of the bulb shall then be immersed for its entire length in an agitated liquid bath at 100° ±0.5°C The resistance of the bulb shall then be measured. The resistance of the bulb shall not differ by more than 1.3 ohms from the resistance at the same temperature when measured without cooling the bulb head.

* 4.5.3.4 Thermoelectric potential. The potentials produced in the bulb winding measured with a potentiometer, when the sensitive element of the bulb is subjected to an air temperature of 150° ±5°C shall not exceed 1 millivolt.

* 4.5.3.5 Self heating. The bulb shall be immersed to the top of the attaching threads in an agitated bath of water at a temperature of 25°C. A current of 150 milliamperes shall be applied to the bulb and the rise in resistance measured. The rise shall not exceed 7 ohms.

* 4.5.3.6 Low temperature exposure. The bulb shall be subjected to a temperature of -65° ±2°C for a period of 48 hours. With the bulb at -65° ±2°C it shall be checked for continuity. After the bulb has returned to room temperature, it shall be subjected to the test specified for bulb adjustment. There shall be no damage which would adversely affect subsequent operation.

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* 4.5.3.7 High temperature exposure. The sensitive element of the bulb shall be subjected to a temperature of 300° ±1°C for a period of 15 minutes. After the bulb has returned to room temperature, it shall be subjected to the test specified for bulb adjustment. There shall be no damage which would adversely affect subsequent operation.

* 4.5.3.8 Vibration endurance. The bulb shall be mounted, by the attaching threads, on a suitably designed vibration stand as specified herein. No additional support of the stem shall be made other than that provided by the attaching threads. The simple harmonic motion shall be applied perpendicular to the longitudinal axis of the bulb stem. The bulb shall be vibrated at 700 cycles per second and 0.003-inch double amplitude for 10 million cycles. For at least the last 15 minutes of the test the bulb element and insulation shall be monitored as in 4.5.1.3. At the conclusion of the test the bulb shall meet the same requirements as 4.5.1.3. No other damage or failure of the bulb shall occur as a result of this test.

* 4.5.3.9 Humidity. The bulb shall be subjected to humidity tests, Procedure I, of MIL-E-5272, except that the temperature shall be +65° ±2°C in a period of 10 cycles. There shall be no evidence of corrosion or rusting of the bulb or any other damage as a result of this test.

* 4.5.3.10 Insulation resistance. Immediately following the humidity test, a potential of 100 volts shall be applied between either electrical receptacle pin and the bulb housing. Air drying of the bulb receptacle, connecting leads and outer housing, prior to conducting this test, is permissible. The insulation resistance of the bulb shall be not less than 20 megohms.

* 4.5.3.11 Salt spray. The bulb shall be subjected to a salt spray test, Procedure I of MIL-E-5272 for a period of 50 hours. At the end of the 50-hour period, the bulb shall be subjected to the tests specified in 4.5.2.1 and 4.5.3.10. No damage to the bulb shall result from this test.

* 4.5.3.12 Internal examination. At least one bulb shall be disassembled and shall comply with 3.4.3.1.

* 4.5.4 Rejection and retest. When tests are specified on a quantity of bulbs that are selected as representative of a certain lot, and two or more of this number fail to meet the requirements, the lot shall be rejected. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the bulbs.

* 4.6 All parts, specimens, or assemblies destroyed in making tests required by this specification or drawings, to determine compliance with the specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

* 4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

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5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.4).

* 5.1.1 Level A. Each bulb shall be preserved and packaged in accordance with MIL-P-116, Method IC1 (no preservative) and placed in a snug-fitting container conforming to PPP-B-676.

* 5.1.2 Level C. Each bulb shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it meets this requirement.

* 5.2 Packing. Packing shall be level A, B or C as specified (see 6.4).

* 5.2.1 Level A. Bulbs preserved and packaged as specified in 5.1.1 shall be packed in weather resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall comply to the applicable special requirements table of the specification. Interior cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the container specification. Containers shall be of uniform shape and size and of minimum cube and tare.

* 5.2.2 Level B. Same as level A, except that waterproofing is not required and domestic containers may be used.

* 5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

* 5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type G-1 bulb covered by this specification is intended for use on aircraft where high-frequency vibration characteristics are present and temperatures in the range of -70°C to +150°C are to be measured, such as carburetor and oil temperatures.

6.2 Definition.

6.2.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from the extreme to the opposite extreme.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: SANE, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

6.4 Ordering data. Levels of packaging and packing required (see Section 5).

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Navy - AS
Air Force - 82

Review Activities:
Army - EL
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Preparing Activity:
Air Force - 82

Project No. 6685-0323

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

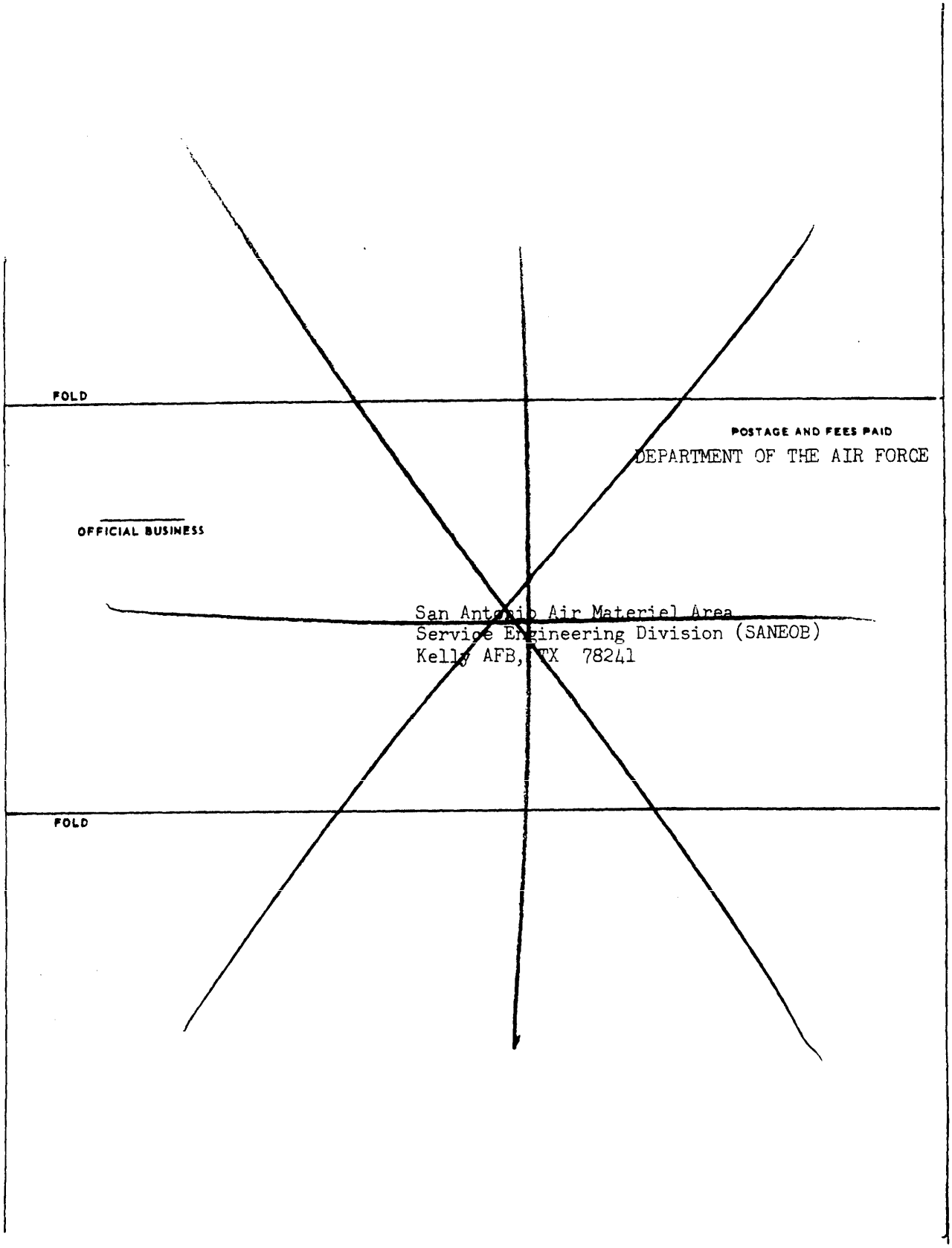
☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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